Chapter 2 Exam Questions

1. Which of the following can implement computing virtualization and improve resource utilization in the era of cloud computing 1.0?
   A. KVM
   B. OpenStack
   C. Hyper-V
   D. Xen

2. （T or F）The virtualization technology can virtualize a physical server into multiple VMs, improving the hardware performance of the physical server.

3. （Multiple）Which of the following statements about Virt-manager are true?
   A. Virt-manager is a graphical management tool for KVM VMs.
   B. Virt-manager is a KVM VM management command set.
   C. Virt-manager is an open-source KVM VM management tool.
   D. Virt-manager is a commercial KVM VM management tool.

4. (Multiple）Which of the following are true about KVM virtualization?
   A. The host OS can be Linux.
   B. The host OS can be Windows.
   C. The guest OS can be Linux.
   D. The guest OS can be Windows.

5. (Single）The KVM virtualization architecture includes the KVM module, QEMU, and Libvirt. Which of the following statements about these modules is false?
   A. The KVM module is responsible for creating vCPUs and allocating virtual memory.
   B. The QEMU module is responsible for simulating I/O operations.
   C. Libvirt and QEMU can form a virtualization solution.
   D. KVM and QEMU can form a virtualization solution.

6. （**T** or F）Virtualization features include partitioning, isolation, encapsulation, and independence.

7. (T or **F**）Virtualization includes computing virtualization, storage virtualization, and network virtualization, which correspond to elastic cloud servers (ECSs), Elastic Volume Service (EVS) disks, and cloud networks in cloud computing, respectively.

8. Which is an open source virtualization technology?
   A. KVM
   B. Fusion compute
   C. Hyper-V
   D. V-sphere

9. (**T** or F) A VM has a configuration file that record the specification and the name of VM.

10. Which is computing virtualization?

    A. I/O virtualization
    B. CPU virtualization
    C. Memory virtualization
    D. Network virtualization

11. The main reason of rapid development in virtualization is because VM outperform the physical machine and has wider applications. **(False)**

12. The main goal of developing virtualization is.
    A. Make full use and utilization of hardware resources.

13. The part of FusionCompute play a role as **a host machine** is.
    a. CNA
    b. VRM
    c. KVM
    d. Libvirt

14. The hypervisor used to monitor the status of host OS. **(False)**

15. In virtualization, only the hypervisor (VMM) can access the hardware resources. **(False)**

16. Which are used to allocate memory and CPU to VM in virtual environment?
    A. Hypervisor
    B. Guest OS
    C. Host OS
    D. VMM

17. If one VM infected by virus or get fault, the other VM's will not affected, which virtualization feature reflect this situation?
    A. Isolation
    B. Encapsulation
    C. Independence
    D. Partition

18. VM consist two files: configuration and data file, this allow migrating or moving the VM to another host for running on it, the features reflect this are.
    A. Isolation
    B. Encapsulation
    C. Independence
    D. Partition

19. The best advantage of virtualization is?
    a. VM migration between hosts
    b. Multiple VMs can run concurrently on a physical host
    c. Multiple VMs can run concurrently on a host OS
    d. In virtualization, The cpu utilization will be 60%

20. Cloud computing and virtualization are inseparable from each other. **False**

21. Open source cloud tools need an open source virtualization technology. **False**

22. Which is true about KVM?

    a. KVM is an open source
    b. KVM is type 2
    c. KVM is module in Linux kernel
    d. KVM is type1

23. Which are true about Xen and KVM?
    a. KVM and Xen are open source virtualization solution
    b. Domain 0 and Domain U are concepts in KVM
    c. KVM and Xen can run in the same host
    d. Xen VM can be converted to KVM VM

24. Which of the following are a KVM management tools?
    A. Libvirt
    B. Virsh command set
    C. Virt-manager
    D. QEMU

25. Which are operations in FusionCompute?
    A. Create VM
    B. Delete VM
    C. Clone VM
    D. Convert external VM to FusionCompute VM

26. Tools that used to create VMs in FusionCompute are
    A. Using GUI
    B. Running commands

27. Which are true about Libvirt management tool?
    A. Is an open source management tool
    B. Is an API developed using C language
    C. Can be called to operate VMs of another languages
    D. Is a part of KVM architecture